

**Department of Software Engineering (SE)**
**Faculty of Science and Information Technology (FSIT)**
**Daffodil International University (DIU)**

**SE331**
**Software Engineering Design Capstone Project**
**Fall 2026**

# Proposed Project Title: MicroJobs

| Group No:02, Lab Section: C1 | |
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**Submission Date:02/10/26**

## Project Overview:

MicroJobs is a student-focused online marketplace where individuals, local shops and small businesses can post short, small-budget tasks and university students can apply, get hired and earn money. Employers publish jobs with a budget and deadline, students submit proposals, and the employer hires the best fit. The platform supports in-app messaging, job completion proof, ratings and reviews, portfolios and user verification so that both sides can trust each other. Students can track their income through an earning dashboard, while admins moderate users, jobs and reports from a central dashboard. The system will be built as a full-stack web application using Next.js and MongoDB.

## Project Stakeholders:

Students / Freelancers (job seekers), Employers / Job Posters (individuals, local shops, small businesses and startups), Platform Administrators, the University / Student Community, and the Project Team (developers and course supervisor).

## Business Need:

Many university students want to earn while studying and gain real work experience, but large freelancing platforms are highly competitive and difficult for beginners. At the same time, small businesses and individuals often need quick, low-cost help with tasks such as poster design, data entry, content writing or simple websites, and have no trusted place to find local student talent. MicroJobs bridges this gap with a simple, trusted platform made for small jobs, where verification, portfolios, reviews and completion proof reduce fraud and build trust, while students build experience, a portfolio and a source of income.

## Tech Stack:

- Language: TypeScript (JavaScript)
- Frontend Framework: Next.js (React, App Router)
- Backend framework: Next.js API Routes / Route Handlers (Node.js)
- Database: MongoDB (MongoDB Atlas)
- Styling: Tailwind CSS
- ORM: Mongoose (ODM for MongoDB)
- Deployment: Vercel (application), MongoDB Atlas (database)
- Any other you want to include: NextAuth.js (authentication), Cloudinary (image/file uploads), Pusher / Socket.io (real-time messaging), Zod (validation), Git & GitHub (version control)

# Functional requirements:

**Module 1: User Management & Trust**

1. **Registration & Login (Shakibur Rahman Akash):** Students and employers create secure accounts (name, email, phone, password, role) and log in with role-based access.
2. **User Verification (Maruf Hasan All Haddy):** Users verify via email, phone or university ID to reduce fake accounts; verified students receive a "Verified Student" badge.
3. **Portfolio Showcase (Shibashis Das):** Students display previous projects, certificates, images, links and work samples for employers to review.
4. **Admin Dashboard (Shakibur Rahman Akash):** Admins manage users, jobs, reports, disputes and platform statistics from one dashboard.

**Module 2: Job Marketplace**

1. **Post a Job (Maruf Hasan All Haddy):** Employers publish small tasks with title, description, required skills, budget, deadline and location.
2. **Apply for Jobs (Shibashis Das):** Students browse and search available jobs and apply with a short message, proposed price and delivery time.
3. **Job Proposal System (Shakibur Rahman Akash):** Students submit customized proposals describing their approach, expected price, delivery time and suitability.
4. **Hire Students (Maruf Hasan All Haddy):** Employers review applications, proposals and profiles, then select and hire a suitable student.

**Module 3: Communication, Delivery & Earnings**

1. **In-App Messaging (Shibashis Das):** Hired students and employers communicate inside the platform to discuss requirements and share task details.
2. **Job Completion Proof (Shakibur Rahman Akash):** Students submit files, screenshots, links or documents as proof when finishing a task, for the employer to review and approve.
3. **Ratings & Reviews (Maruf Hasan All Haddy):** After a job is completed, both sides give star ratings and written reviews to build trust.
4. **Earning Dashboard (Shibashis Das):** Students view total earnings, pending payments, completed-job earnings, withdrawals and payment history.

# Non-functional Requirements:

1. **Performance:** Pages should load within 3 seconds under normal network conditions, and API responses should complete within 1–2 seconds.
2. **Security:** Passwords are hashed (bcrypt), sessions use secure JWT/cookies, all traffic uses HTTPS, and inputs are validated to prevent injection and XSS attacks.
3. **Usability:** The interface is simple, clean and easy to use for first-time users, with a consistent design.
4. **Responsiveness:** The application works smoothly on mobile, tablet and desktop screens.
5. **Scalability:** The system can handle a growing number of users and jobs using MongoDB Atlas and serverless deployment on Vercel.
6. **Reliability & Availability:** The platform targets 99% uptime with regular database backups.
7. **Maintainability:** Code follows a modular structure with TypeScript, reusable components and version control using Git/GitHub.
8. **Data Privacy:** Personal information (phone, ID documents) is visible only to authorized users and admins.